INCH-POUND
MIL-M-38510/51E
AMENDMENT 3
28 APRIL 1999
SUPERSEDING
AMENDMENT 2
31 August 1988

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, FLIP-FLOPS AND LATCHES, MONOLITHIC SILICON

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/51E, dated 30 April 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, add new case outline as follows:

"N	F-5A (16-lead, .440" x .285" x .115"), flat package
T	F-2A (14-lead, .390" x .260" x .115"), flat package".

PAGE 4

4.2a: Delete "3.1.9 through 3.1.13", and substitute "Initial (pre burn-in) electrical parameters (3.1.10) through interim (post burn-in) electrical parameters (3.1.14)".

PAGE 7

Table I, t_{PHL}, device types 01, 02, 51, and 52. Change min limits to 13. Add device types 03 and 53 with min limits of 10 and max limits of 370.
Table I, t_{PLH}, device types 01, 02, 51, and 52. Change min limits to 13. Add device types 03 and 53 with min limits of 10 and max limits of 245.
Table I, t_{PHL} (R) or (S), and t_{PLH} (R) or (S), device types 01, 02, 51, and 52. Change min limits to 13. Device types 03 and 53, change min limits to 10.
Table I, t_{THL}, device types 03, 51, 52, and 53. Change min limits to 10.

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Table I, t_{THL} and t_{TLH}, device types 03, 51, 52, and 53. Change min limits to 10.
Table I, Set-up times: Delete "225" from maximum column, and insert "225" into minimum column, for limits.
Table I, Hold times: Delete "225" from maximum column, and insert "225" into minimum column, for limits.

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AMSC N/A
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Table II, Line 9, insert “10 and 11” in column for class B. Delete line 12 and substitute:

“

12	Group D end point electrical parameters method 5005	4.4.4	1, 2, 3			4.4.4	1, 2, 3			

“

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FIGURE 1: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y” for device types 01 and 51. Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z” for device types 02, 03, 52, and 53.

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FIGURE 5, note 1: Delete “2 kΩ or 47 kΩ”, and substitute “2 kΩ to 47 kΩ”.

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Table III, test 39, pin 1: Add “3/” to I_{OH}.
Table III, test 39, pin 4: Add “7/” to V_{IL1}.
Table III, test 39, pin 6: Add “4/” to V_{IH1}.
Table III, test 47, pin 4: Add “8/” to V_{IL2}.
Table III, test 47, pin 6: Add “5/” to V_{IH2}.
Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”
Table III, test 51, pin 1: Add “6/” to I_{OL}.
*TABLE III, I_{IH1}, Max limits column: change “8.0” to “800.0”.
*TABLE III, I_{IH2}, Max limits column: change “1.0” and “45.0” to “100.0” and “100.0” respectively.

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”
*TABLE III, I_{IL1}, Max limits column: change “-8.0” to “-800.0”.
*TABLE III, I_{IL2}, Max limits column: change “-1.0” and “-45.0” to “-100.0” and “-100.0” respectively.

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”
Table III, t_{PHL}, t_{PHL} (R) or (S), t_{PLH}, t_{PLH} (R) or (S): Change subgroup 9 min test limits to 13 ns, change subgroup 10, min test limits to 18 ns, change subgroup 11, min test limits to 13 ns.
Table III, t_{THL} and t_{TLH}: Change subgroup 9 min test limits to 10 ns, change subgroup 10 min test limits to 14 ns.

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, test 35, pin 1: Add “3/” to I_{OH}.
Table III, test 35, pin 4: Add “7/” to I_{IL1}.
Table III, test 35, pin 7: Add “4/” to V_{IH1}.
Table III, test 43, pin 4: Add “8/” to V_{IL2}.
Table III, test 43, pin 7: Add “5/” to V_{IH2}.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, test 47, pin 1: Add “6/” to I_{OL}.
*TABLE III, I_{IH1}, Max limits column: change “10” to “1000.0”.
*TABLE III, I_{IH2}, Max limits column: change “1.0” and “45.0” to “100.0” and “100.0” respectively.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
*TABLE III, I_{IL1}, Max limits column: change “-10” to “-1000.0”.
*TABLE III, I_{IL2}, Max limits column: change “-1.0” and “-45.0” to “-100.0” and “-100.0” respectively.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, t_{PHL}, t_{PHL} (R) or (S), t_{PLH}, t_{PLH} (R) or (S): Change subgroup 9 min test limits to 13 ns, change subgroup 10 min test limits to 18 ns, change subgroup 11 min test limits to 13 ns.
Table III, t_{THL} and t_{TLH}: Change subgroup 9 min test limits to 10 ns, change subgroup 10 min test limits to 14 ns.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III: tests 163 and 164: Delete “300, 450, and 300” for minimum column, and insert “300, 450, and 300” into maximum column, for test limits.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, test 25, pin 2: Add “3/” to I_{OH}.
Table III, test 25, pin 3: Add “7/” to V_{IL1}.
Table III, test 25, pin 4: Add “4/” to V_{IH1}.
Table III, test 33, pin 3: Add “8/” to V_{IL2}.
Table III, test 33, pin 4: Add “5/” to V_{IH2}.
Table III, test 37, pin 2: Add “6/” to I_{OL}.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, t_{PHL} (R): Change subgroup 9 min test limits to 10 ns, change subgroup 10 min test limits to 14 ns, change subgroup 11 min test limits to 10 ns.
Table III, t_{PLH}: Change subgroup 10 min test limits to 10 ns. T_{THL}: Change subgroup 10 min test limits to 14 ns, change subgroup 11 min test limits to 10 ns. T_{THL}: Change subgroup 9 min test limits to 10 ns, change subgroup 11 min test limits to 10 ns.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”

PAGES 45 AND 46

Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”

PAGE 47

Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”
*TABLE III, I_{IH1}, Max limits column: change “8.0” to “800.0”.

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”
*TABLE III, I_{IH2}, Max limits column: change “1.0” and “45.0” to “100.0” and “100.0” respectively.
*TABLE III, I_{IL1}, Max limits column: change “-8.0” to “-800.0”.
*TABLE III, I_{IL2}, Max limits column: change “-1.0” and “-45.0” to “-100.0” and “-100.0” respectively.

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”
Table III, t_{PHL}, t_{PHL} (R) or (S), t_{PLH}, and t_{PLH} (R) or (S): Change subgroup 9 min test limits to 13 ns, change subgroup 10 min test limits to 18 ns, change subgroup 11 min test limits to 13 ns.
Table III, t_{THL} and t_{TLH}: Change subgroup 9 min test limits to 10 ns, change subgroup 10 min test limits to 14 ns, change subgroup 11 min test limits to 10 ns.

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Table III: Delete “Cases A, C, D, X, and Y” and substitute “Cases A, C, D, T, X, and Y.”

PAGE 51 AND 52

Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”

PAGE 53

Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
*TABLE III, I_{IH1}, Max limits column: change “10” to “1000.0”.
*TABLE III, I_{IH2}, Max limits column: change “1.0” and “45.0” to “100.0” and “100.0” respectively.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
*TABLE III, I_{IL1}, Max limits column: change “-10” to “-1000.0”.
*TABLE III, I_{IL2}, Max limits column: change “-1.0” and “-45.0” to “-100.0” and “-100.0” respectively.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, t_{PHL}, t_{PHL} (R) or (S), t_{PLH}, and t_{PLH} (R) or (S): Change subgroup 9 min test limits to 13 ns, change subgroup 10 min test limits to 18 ns, change subgroup 11 min test limits to 13 ns.
Table III, t_{THL} and t_{TLH}: Change subgroup 9 min test limits to 10 ns, change subgroup 10 min test limits to 14 ns, change subgroup 11 min test limits to 10 ns.

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Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, tests 211 and 212: Delete “300, 450, and 300” from minimum column, and insert “300, 450, and 300” into maximum column, for test limits.

PAGE 57 THROUGH 59

Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”

PAGE 60

Table III: Delete “Cases E, F, and Z” and substitute “Cases E, F, N, and Z.”
Table III, t_{PHL} (R): Change subgroup 9 min limits to 10 ns, change subgroup 10 min test limits to 14 ns, change subgroup 11 min test limits to 10 ns.
Table III, t_{PLH}: Change subgroup 10 min test limits to 10 ns. T_{THL}: Change subgroup 10 min test limits to 14 ns, change subgroup 11 min test limits to 10 ns. T_{TLH}: Change subgroup 9 min test limits to 10 ns, change subgroup 11 min test limits to 10 ns.

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4.4.1b: Delete “5 and 6”, and substitute “5, 6, and 8”.
4.4.1, add the following new paragraph:

“d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance”.
“e. When device types 01 through 03 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 53, respectively.”

4.4.3: Delete 4.4.3c and substitute as follows:

“c When device types 01 through 03 are qualified by extension (see 4.3.1), these device types will be inspected (QCI) according to the requirements for device types 51 through 53, respectively.”

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Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Army - AR, MI, SM
Air Force - 19, 85, 99

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